

P&ID Layer Structure Documentation

Both improved sheets use an identical, strict **7-level Inkscape layer**

hierarchy. Layers are authored bottom-to-top so that later layers draw above earlier ones. Each layer carries an `inkscape:groupmode="layer"` attribute and a human-readable `inkscape:label`, so the file opens with named, toggleable layers in Inkscape / Illustrator.

SVG (A3 landscape, viewBox 0 0 1587.273 1122.430)

-  `<defs>` markers / gradients carried **from** source
-  L1 Border / Title block id = layer-frame
 -  white sheet background
 -  **double** drawing border
 -  structured title block (project / title / dwg **no** / standard / rev)
 -  sheet header strip
-  L2 Equipment & Vessels id = layer-equipment
 -  re-emitted structural geometry (vessel/tank/coupler outlines, frames)
 -  standard equipment glyphs (vessels, cavities, couplers, HX, terminal points, pickup antennae, nodes) placed **from** segmentation
-  L3 Process Lines id = layer-process
 -  L3.A id = layer-line-A **Class** A 4.5 K / 3 bar (blue)
 -  L3.B id = layer-line-B **Class** B 3.5 K / 27 mbar (teal)
 -  L3.D id = layer-line-D **Class** D 40 K / 14 bar (red)
 -  L3.E id = layer-line-E **Class** E 60 K / 13 bar (olive)
 -  L3.WATER id = layer-line-WATER DI cooling water (green)
 -  L3.QINFRA id = layer-line-QINFRA **scope** division (dk green dash)
 -  L3.AIR id = layer-line-AIR instrument air (magenta d-dot)
 - (**within each class**  fills  junction nodes  lines, drawn **in** that **order**)
-  L4 Vacuum Barriers & Boundaries id = layer-vacuum
 -  dashed vacuum-vessel / **scope**-division boundaries
 -  "vacuum barrier" annotations
-  L5 Instrumentation Symbols id = layer-instruments
 -  ISA-5.1 instrument bubbles (circle; dashed = protection/safety)
 -  valve bodies (hand / control / solenoid / relief)
 -  family **fill**: white=LB, salmon=RFCELL, light-blue=LBI
-  L6 ISA 5.1 Tags & Labels id = layer-tags
 -  two-line bubble tags ([variable][function] **over** [loop])
 -  valve tag labels
-  L7 Legend & Annotations id = layer-legend
 -  **right column**  instrument / valve / equipment symbol **key** + tag grammar
 - bottom **band**: process-line **class** legend (colour T/P service)

Layer purpose summary

Layer	id	Purpose	Source of content
L1	layer-frame	Sheet, border, title block	Generated
L2	layer-equipment	Vessels / equipment & structure	Re-emitted structure + segmentation equipment
L3	layer-process (+ 7 class sub-layers)	Colour-segmented process piping	Re-extracted source geometry, recoloured
L4	layer-vacuum	Vacuum / scope boundaries	Re-extracted dashed boundaries + segmentation
L5	layer-instruments	ISA bubbles & valves	Segmentation instrument/safety coordinates
L6	layer-tags	ISA-5.1 tags & labels	Segmentation tags
L7	layer-legend	Legend & annotations	Generated

Process-line class → layer mapping

Sub-layer id	Class	Service	Colour	Style
layer-line-A	A	4.5 K / 3 bar — LHe supply	#0033cc	solid
layer-line-B	B	3.5 K / 27 mbar — 2 K LP return	#00a6bd	solid
layer-line-D	D	40 K / 14 bar — thermal shield	#e00000	solid
layer-line-E	E	60 K / 13 bar — return / guard	#8a8a00	solid
layer-line-WA- TER	Water	DI cooling water	#00a000	solid
layer-line-QIN- FRA	QINFRA	scope division / infrastructure	#006400	dashed
layer-line-AIR	Inst. air	pneumatic 6 (5- 7) bar(g)	#c000c0	dash-dot

Toggle any class sub-layer in Inkscape to isolate a single cryogenic service.